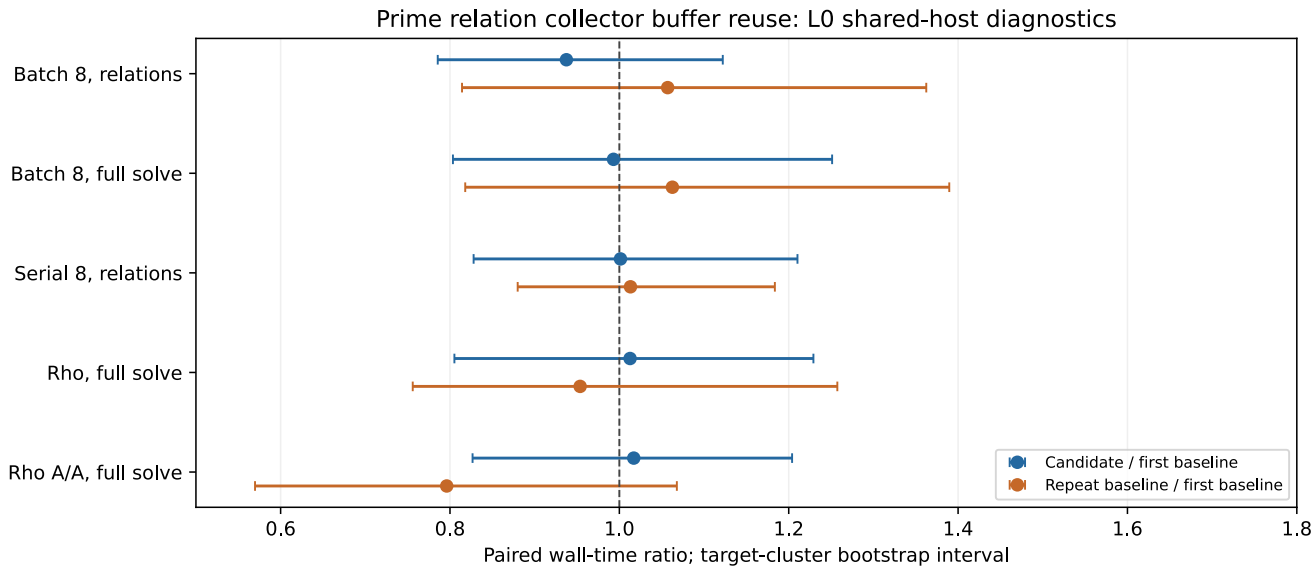
Reusing prime lane collector buffers

The prime relation collector now reuses four vectors between synchronized lane rounds: requests, native left and right points, and outputs. The release stream test passed at widths 4, 8, and 16. The saved baseline and candidate expanded to identical public-target benchmark plans.



The chart shows L0 wall-time ratios on one shared macOS arm64 host. Each point is a geometric mean over matched measured rounds; whiskers are target-cluster bootstrap intervals over eight public targets. Blue compares the candidate with the first baseline. Orange compares the repeated baseline with the first baseline. The dashed line is parity.

Results and scope

The three sealed sessions contain 384/384 verified executions and 288/288 identical measured replays. The corrected cross-session checker found zero mismatches among compared deterministic fields in 128 candidate/baseline record pairs and 128 baseline/baseline pairs. All IC arms had mean cold counted cost 7,952.58 GAE; sealed native field work remains unpriced. The native ecbench comparison reports counted-cost ratio 1.000 over 24 measured batch-8 pairs and refuses a cross-session wall comparison. In the candidate session, batch-8 IC / matched rho has lower-bound counted-cost ratio 6.8747 (interval 5.6689–8.4198).

Custom L0 diagnostics: batch-8 relation-phase wall ratio 0.937 (95% target-cluster interval 0.786–1.122) and whole-solve wall ratio 0.993 (0.803–1.251). The repeated baseline ratios were 1.057 (0.814–1.362) and 1.063 (0.818–1.390), respectively. Every interval includes parity. The run verifies unchanged deterministic behavior and the source-level buffer reuse, while the L0 timings do not establish a wall-time improvement.

The baseline binary SHA-256 is `ab5764921ad616ae8177ca57c125ec8cd7ba09896f1e0644a6999d1fb33b7159`, bound to source commit `431d0b3167027092443705e21d363489b2637a22` by the prior postmerge build receipt. The candidate binary SHA-256 is `db9cf52602e8b3bc19a885c3eae99c899bd6acc541958246f3e540e58e8de743`, with session source commit `316cc8386611d9b6ba7a57c86442b4d59ba9ee58`.

The first release build failed on temporary-storage exhaustion and the first comparison checker classified phase wall time as a deterministic mismatch. Both failed outputs and their corrections are retained. Raw sessions, replay audits, comparison JSON, protocol, and reproduction commands are linked from `REPORT.md`.